Analytical Chemistry Cumulative Exam

September 9, 2011

Sampling

1. Define "sampling."

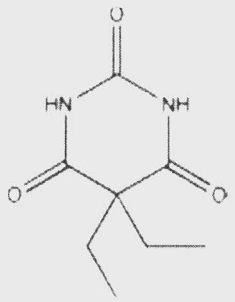
2. Define "sample preparation." Be sure to broadly cover all necessary steps for a sample preparation in your definition.

3. (Derived from Harris, 7th Edition, Problem 28-12) From the standard reduction potentials, which of the following metals (given below) would theoretically dissolve in HCl by the following reaction where "M" is a metal and "*n*" is a stoichiometric coefficient:

$$M + nH^+ \rightarrow M^{n+} + n/2\ H_2\ (g)$$

Metals: Al, Au, Co, Pt, and Zn

The reduction potentials are as follows:

$2H^+ + 2e^- \leftrightarrows H_2(g)$	$E^\circ = 0.000$
$Cl_2\ (aq) + 2e^- \leftrightarrows 2Cl^-$	$E^\circ = 1.396$
$Al^{3+} + 3e^- \leftrightarrows Al(s)$	$E^\circ = -1.677$
$Au^+ + e^- \leftrightarrows Au(s)$	$E^\circ = 1.69$
$Co^{2+} + 2e^- \leftrightarrows Co(s)$	$E^\circ = -0.282$
$Pt^{2+} + 2e^- \leftrightarrows Pt(s)$	$E^\circ = 1.18$
$Zn^{2+} + 2e^- \leftrightarrows Zn(s)$	$E^\circ = -0.762$

4. An analysis requires isolation of barbital from urine (see structure below). Barbital has a reported pK_a of 7.95 and a log *P* of 0.69. Provide a workflow for the isolation of barbital from urine such that an analysis can be performed by LC-MS.